

Module 8 : Matched pair randomized experiments

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Design of Experiments - Stat 140

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Matched pair design

- Special case of blocked design.
- Randomized experiments with, e.g., "identical" twins, similar classrooms within a school, similar hospitals within a city, similar cities within a country. SUTVA?
- Often more "efficient" than a completely randomized experiment.
- Can be useful for the analysis of observational studies.

Assignment mechanism

- Each unit has probability $\frac{1}{2}$ of being assigned to the active treatment.
- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y}(0),\mathbf{Y}(1)) = \begin{cases} 2^{-N/2} & \text{if } \mathbf{W} \in W^+ \\ 0 & \text{otherwise.} \end{cases}$$

Science Table

Unit i	Pair p	$Y_{i,p}(0)$	$Y_{i,p}(1)$
1	1	$Y_{1,1}(0)$	$Y_{1,1}(1)$
2	1	$Y_{2,1}(0)$	$Y_{2,1}(1)$
3	2	$Y_{3,2}(0)$	$Y_{3,2}(1)$
4	2	$Y_{4,2}(0)$	$Y_{4,2}(1)$
...	...	...	...
...	...	...	...
$N-1=2*P-1$	P	$Y_{N-1,P}(0)$	$Y_{N-1,P}(1)$
$N=2*P$	P	$Y_{N,P}(0)$	$Y_{N,P}(1)$

- Causal estimands

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...	...	...	...
...	...	...	...
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$N=2*P$	P	$Y_{N,P}(0)$	$Y_{N,P}(1)$

- Causal estimands

- Unit-level causal effects : $\tau_{i,p} = Y_{i,p}(1) - Y_{i,p}(0)$

Unit i	Pair p	$Y_{i,p}(0)$	$Y_{i,p}(1)$
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3	2	$Y_{3,2}(0)$	$Y_{3,2}(1)$
4	2	$Y_{4,2}(0)$	$Y_{4,2}(1)$
...	...	...	...
...	...	...	...
$N-1=2*P-1$	P	$Y_{N-1,P}(0)$	$Y_{N-1,P}(1)$
$N=2*P$	P	$Y_{N,P}(0)$	$Y_{N,P}(1)$

- Causal estimands

- Unit-level causal effects : $\tau_{i,p} = Y_{i,p}(1) - Y_{i,p}(0)$
- Pair-level causal effects : $\tau_p = \frac{1}{2} \sum_{i \in \text{Pair } p} [Y_{i,p}(1) - Y_{i,p}(0)]$

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...	...	...	...
...	...	...	...
$N-1=2*P-1$	P	$Y_{N-1,P}(0)$	$Y_{N-1,P}(1)$
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- Causal estimands

- Unit-level causal effects : $\tau_{i,p} = Y_{i,p}(1) - Y_{i,p}(0)$
- Pair-level causal effects : $\tau_p = \frac{1}{2} \sum_{i \in \text{Pair } p} [Y_{i,p}(1) - Y_{i,p}(0)]$
- Average causal effect : $\tau = \frac{1}{N} \sum_{i=1}^N [Y_{i,p}(1) - Y_{i,p}(0)] = \frac{1}{P} \sum_{p=1}^P \tau_p$

Observed Table

Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	0	$Y_{1,1}(0)$	?	$Y_{1,1}^{obs}$
2	1	1	?	$Y_{2,1}(1)$	$Y_{2,1}^{obs}$
3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

Fisherian inference

Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	0	$Y_{1,1}(0)$	?	$Y_{1,1}^{obs}$
2	1	1	?	$Y_{2,1}(1)$	$Y_{2,1}^{obs}$
3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

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3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

- Test statistic

- $\bar{Y}_t^{obs} - \bar{Y}_c^{obs} = \frac{1}{P} \sum_{p=1}^P [Y_{p,t}^{obs} - Y_{p,c}^{obs}]$
- T_{paired}
- $T_{\text{rank,pair}}$

Neymanian inference

Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	0	$Y_{1,1}(0)$	?	$Y_{1,1}^{obs}$
2	1	1	?	$Y_{2,1}(1)$	$Y_{2,1}^{obs}$
3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

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3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

- Estimator for τ_p :

Neymanian inference

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1	1	0	$Y_{1,1}(0)$	?	$Y_{1,1}^{obs}$
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3	2	0	$Y_{3,2}(0)$	?	$Y_{3,2}^{obs}$
4	2	1	?	$Y_{4,2}(1)$	$Y_{4,2}^{obs}$
...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
N	P	1	?	$Y_{N,P}(1)$	$Y_{N,P}^{obs}$

- Estimator for τ_p :

- $$\hat{\tau}_p = Y_{p,t}^{obs} - Y_{p,c}^{obs} = \sum_{i \in \text{Pair } p} (2W_{i,p} - 1) Y_{i,p}^{obs}$$

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Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	0	$Y_{1,1}(0)$	?	$Y_{1,1}^{obs}$
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...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
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- Estimator for τ_p :

- $$\hat{\tau}_p = Y_{p,t}^{obs} - Y_{p,c}^{obs} = \sum_{i \in \text{Pair } p} (2W_{i,p} - 1) Y_{i,p}^{obs}$$

- Sampling variance of $\hat{\tau}_p$:

Neymanian inference

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...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
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- Estimator for τ_p :
 - $\hat{\tau}_p = Y_{p,t}^{obs} - Y_{p,c}^{obs} = \sum_{i \in \text{Pair } p} (2W_{i,p} - 1) Y_{i,p}^{obs}$
- Sampling variance of $\hat{\tau}_p$:
 - $Var(\hat{\tau}_p) = S_c^2(p) + S_t^2(p) - \frac{S_{ct}^2(p)}{2}$

Neymanian inference

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...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
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- Estimator for τ :

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...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
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- Estimator for τ :

- $$\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs} = \frac{1}{P} \sum_{p=1}^P \hat{\tau}_p$$

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...	...	...	...	...	...
...	...	...	...	...	...
N-1	P	0	$Y_{N-1,P}(0)$	?	$Y_{N-1,P}^{obs}$
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- Estimator for τ :
 - $\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs} = \frac{1}{P} \sum_{p=1}^P \hat{\tau}_p$
- Sampling variance of $\hat{\tau}$:

Neymanian inference

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...	...	...	...	...	...
...	...	...	...	...	...
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- Estimator for τ :
 - $\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs} = \frac{1}{P} \sum_{p=1}^P \hat{\tau}_p$
- Sampling variance of $\hat{\tau}$:
 - $Var(\hat{\tau}) = \frac{1}{P^2} \sum_{p=1}^P [S_c^2(p) + S_t^2(p) - \frac{S_{ct}^2(p)}{2}]$